FIGURE 7.10 Short-term memory is like a hand full of eggs; it can hold only a limited number of items at a time.

as many as 80 digits, as shown in Figure 7.12, by using extraordinary strategies for chunking. He was a competitive runner, so he might store the sequence "3492 . . ." as "3 minutes, 49.2 seconds, a near world-record time for running a mile." He might store the next set of numbers as a good time for running a kilometer, a mediocre marathon time, or a date in history. However, when he was tested on his ability to remember a list of letters, his performance was only average, because he had not developed any special chunking strategies for letters.

As a rule, a psychologist who reads a list of numbers and asks someone to repeat them assumes that the results measure short-term memory. The possibility of chunking, however, shows that the test is not a pure measure of short-term memory. Someone who recognizes 1492 and 1776 as meaningful units is clearly bringing long-term memories into this short-term memory test. In the case of the Carnegie-Mellon student, it is not clear whether he was using short-term memory at all or whether he had actually developed ways to rapidly organize large amounts of information into long-term memory.

Decay of Short-Term Memories over Time People can forget long-term memories, largely through proactive and retroactive interference, but the mere passage of time (decay) does not weaken long-term memories by very much. In contrast, short-term memories can disappear quickly, apparently through simple decay, unless people continually rehearse them.

Lloyd Peterson and Margaret Peterson (1959) demonstrated the decay of short-term memory with an experiment that you can easily repeat if you coax a friend to volunteer. First, read aloud a meaningless sequence of letters, such as HOZDF. Then wait for a bit and ask your friend to repeat it—an easy task, because your friend spent the delay rehearsing, "H-O-Z-D-F, H-O-Z-D-F, . . ."

Forgetting will occur, however, if you prevent rehearsal during the delay. Say "I am going to read you a list of letters, such as HOZDF. Then I'm going to tell you a number, such as 231. When you hear the number, begin counting backward by threes: 231, 228, 225, 222, 219, and so on. When I tell you to stop, I'll ask you to repeat the sequence of letters." Record your data as in Table 7.2.

TRY IT YOURSELF

Try this experiment with several friends, and compute the percentage of those who recalled the letters correctly after various delays. Figure 7.13 gives the results that Peterson and Peterson obtained. Note that only about 10% of their subjects could recall the letters correctly after a delay of 18 seconds. In other words, if we fail to rehearse something that has entered short-term memory, it will generally fade away within 20 seconds or less.

This demonstration breaks down, however, if you use highly meaningful material. If you ask your friend to memorize "there is a poisonous snake under the chair," he or she is likely to remember it well even after counting backward by threes from 231. Highly meaningful material enters long-term memory quickly.

Furthermore, Peterson and Peterson's experiment does not demonstrate such a distinct difference between short-term and long-term memory as we once thought existed. Let's reconsider the demonstration: You read off, say,

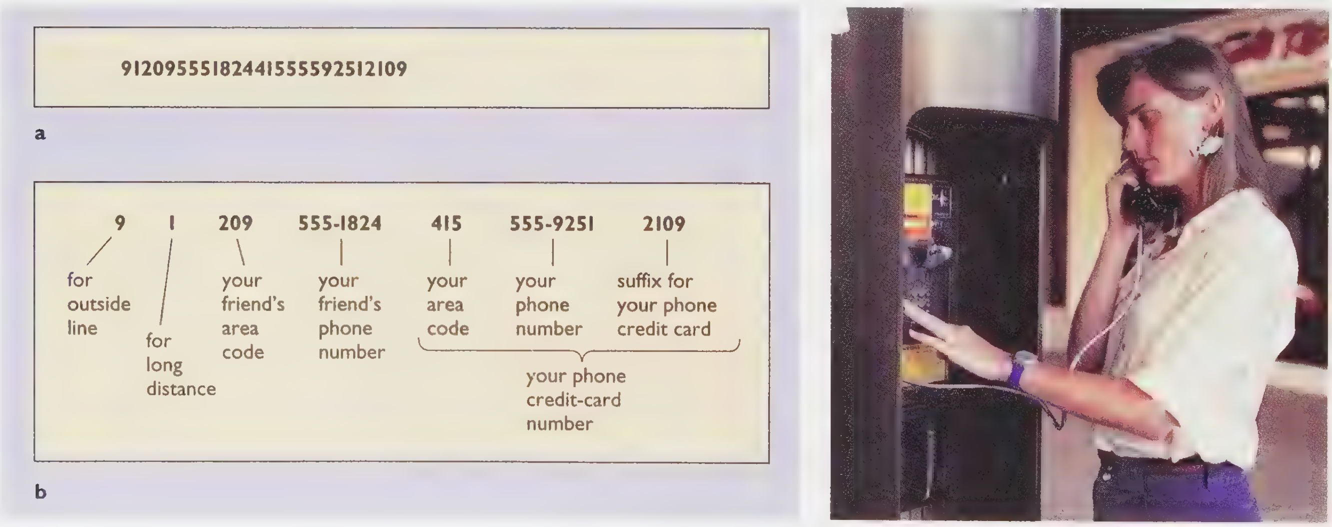


FIGURE 7.11 We overcome the limits of short-term memory through chunking. You probably could not remember the 26-digit number in (a), but by breaking it up into a series of chunks, you can remember it and dial the number correctly.

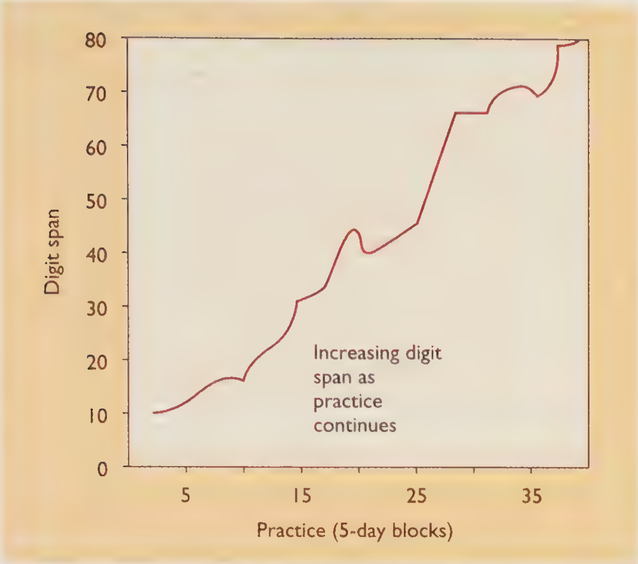


FIGURE 7.12 Most people can repeat a list of about seven numbers. One college student gradually increased his ability to repeat a list of numbers, over 18 months of practice. With practice, he greatly expanded his short-term memory for digits but not for letters or words. (From Ericsson, Chase, & Falcon, 1980.)

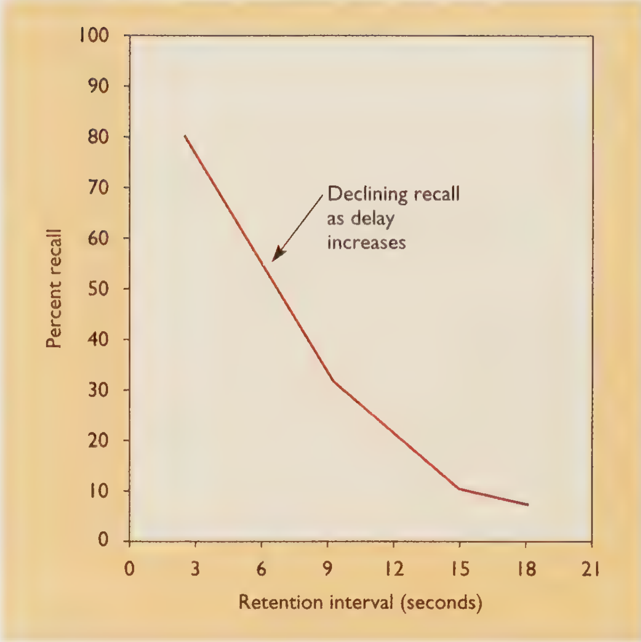


FIGURE 7.13 In a study by Peterson and Peterson (1959), people remembered a set of letters well after a short delay, but their memory faded greatly over 20 seconds if they were prevented from rehearsing during that time.

"BKLRE," asked someone to count backward by threes for 5 seconds, and then probably found that the person could recall the letters. After two more trials you came to "DSJGT" and a 20-second delay, and the person probably could not recall it. But DSJGT was the fourth trial, potentially subject to a fair amount of proactive interference. Suppose you try this with another volunteer, but now you *start* with DSJGT and a 20-second delay. Your new volunteer will probably remember DSJGT (Keppel & Underwood, 1962; Wickens, 1970). Evidently, the forgetting of items stored in short-term memory depends partly on interference, as it does with long-term memory.

CONCEPT CHECK

4. Name one way in which short-term memory and long-term memory are evidently different and one way in which they are similar. (Check your answers on page 240.)

TABLE 7.2

LETTER SEQUENCE	STARTING NUMBER	DELAY IN SECONDS	CORRECT RECALL?
BKLRE	712	5	
ZIWOJ	380	10	
CNVIU	416	15	
DSJGT	289	20	
NFMXS	601	25	

The Transfer from Short-Term Memory to Long-Term Memory

For many years, psychologists thought of short-term memory and long-term memory as separate stages: Information first enters short-term memory and stays there for a while. If rehearsal kept it there long enough, it would **consolidate** to *form a long-term memory* (Figure 7.14).

Now we suspect that consolidation depends on much more than just time and that holding something for a long time in short-term memory does not automatically shift it into long-term memory. As an illustration, try the following experiment (based on Craik & Watkins, 1973): Read the list of words below to yourself, or read them to your roommate.

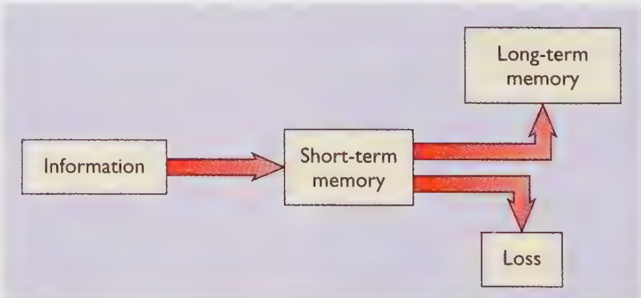


FIGURE 7.14 According to the original conception of the relationship between short-term and long-term memory, if a short-term memory is rehearsed long enough, it becomes a long-term memory; without consolidation, it is lost. This view is now considered oversimplified.

Some of the words start with *g*. The instruction for part 1 of the experiment is to keep track of the most recent *g*-word.

TRY IT YOURSELF

At the end of the list, you (or your roommate) should say what was the last word on the list that began with *g*:

table, giraffe, frog, goose, key, window, banana, pencil, spoon, grass, road, garden, house, tree, lake, paper, chicken, glove, paint, garlic, stone.

The correct answer is *garlic*, and most people easily come up with the right answer. So far, so good. Note that what you needed to do was to store each *g*-word in short-term memory until you dumped it to replace it with the next *g*-word. You had to hold some of the words briefly in memory and others for longer times. Okay, now that I have distracted you for a while, let's move on to part 2 of the experiment: I didn't warn you about this, but I now want you (or your roommate) to write a list of *all* the *g*-words on that list. (Try this before you continue reading.)

The correct answers are *giraffe, goose, grass, garden, glove,* and *garlic*. Considering the delay between when you first read the words and when you recalled them, we must assume you recalled them from long-term memory. The question is, which terms made it into your long-term memory? In general, researchers find that people are about equally likely to remember *giraffe, grass,* and *glove* (which stayed in short-term memory only briefly) as to remember *goose* or *garden* (which stayed longer in short-term memory before being replaced). Evidently, how long a word stays in short-term memory has little to do with whether it moves into long-term memory.

Another problem with the traditional concept: If short-term memory is just a holding station on the way to long-term memory, then any defect in short-term memory should block the formation of long-term memories. However, a few brain-damaged patients can form and retrieve new long-term memories normally, despite serious defects in short-term memory. One such patient had a short-term memory capacity of only *two* items. For example, he might be able to repeat "7, 4" but would fail at repeating "7, 4, 8" (Shallice & Warrington, 1970). However, with extensive repetitions, he could memorize lengthier materials (just as a normal person could memorize something longer than seven items), and after memorizing it, he could retain it over the long term.

It appears to be possible to have a poor short-term memory and still store a great deal into long-term memory. That conclusion implies that short-term memory is not just a holding stage on the way to long-term memory.

A Modified Theory: Working Memory

The concept of short-term memory has a status similar to that of many other psychological concepts: The original concept is partly right, but not quite. Most memory researchers today have substituted a different term, working memory. **Working memory** is seen not just as a system that temporarily stores information, but also as ***a system for processing or working with current information.*** That is, the concept is almost synonymous with one's current sphere of attention. It includes short-term memory (events that just occurred), but it also includes information already stored in long-term memory and now recalled for current use. Working memory includes at least three major components (Baddeley & Hitch, 1994):

- *A central executive,* which governs shifts of attention. When someone tries to manage two tasks at once, the central executive has to allocate resources to the two and determine when to attend to one and when to the other. If all this sounds like a "little person in the head," it is because psychologists have not yet figured out how the central executive operates. Still, people who are good at one kind of divided-attention task tend to be good at others also, and people with damage to the frontal cortex have trouble with all such tasks (Cummings, 1995). So the central executive is a meaningful process, even if we do not yet understand it very well.
- *A phonological loop,* which stores and rehearses speech information. The phonological loop enables us to repeat seven or so numbers or letters immediately after hearing them; it more or less corresponds to the traditional view of short-term memory.
- *A visuospatial sketchpad,* which stores and manipulates visual and spatial information, providing for vision what the phonological loop provides for speech. Research indicates that its limit is about four visual items (Luck & Vogel, 1997).

The reason for distinguishing between the phonological and visuospatial stores is that two word tasks or two visuospatial tasks performed at the same time interfere with each other, but people can perform one of each at the same time without interference (Baddeley & Hitch, 1974; Hale, Myerson, Rhee, Weiss, & Abrams, 1996). People may have working memory stores for touch, smell, and taste also, but so far researchers have concentrated mostly on the auditory and visual stores.

Consistent with the idea of a phonological loop, it is easier to remember a list of short words that are easy to pronounce than a list of longer words that are harder to pronounce (Cowan, Wood, Nugent, & Treisman, 1997), and it is easier to remember words that sound different than words that sound similar to one another (Awh et al., 1996).

TRY IT YOURSELF

For example, try to memorize the following lists. Most will find List 1 the easiest.

List 1 (short words)	List 2 (longer words)	List 3 (similar-sounding words)
fit	prayer	Swiss
blue	yield	stress
tell	crowd	sauce
pin	blouse	sells
trip	strolls	swells
high	thrill	sews
deck	flour	space

The concept of a phonological loop helps to explain differences in reading comprehension. Imagine various people reading either of the following sentences:

> Because Ken really liked the boxer, he took a bus to the nearest pet store to buy the animal.
>
> Because Ken really liked the boxer, he took a bus to the nearest sports arena to see the match.

Then, the readers are asked true-false questions such as

T F Ken liked a dog.
T F Ken liked a fighter.

Many English words have several unrelated meanings—for example, *boxer, bass, tire, leaves.* To understand such words, we must wait until the context of the sentence identifies the correct meaning. In the "Ken" sentences, the context remained ambiguous about the meaning of "boxer" until eight words later—a serious strain for any reader with a short phonological loop (Miyake, Just, & Carpenter, 1994).

CONCEPT CHECK

5. Would readers who have a shorter phonological loop show greater reading comprehension in English, which has many words with two or more meanings, or in Italian, which has few such words? (Check your answer on page 241.)

Other Memory Distinctions

Later in this chapter we shall encounter the distinction between declarative memories and procedural memories. A *declarative memory* is the ability to state a fact; a *procedural memory* is a skill, a memory of how to do something. Certain kinds of brain damage impair declarative memory without damaging procedural memories. That is, a man who forgets his address and telephone number may nevertheless remember how to tie his shoes and how to ride a bicycle.

Psychologists also find it useful to distinguish between two kinds of long-term declarative memory—**semantic memory,** *memory of general principles,* and **episodic memory,** *memory for specific events in a person's life, generally including details of when and where they happened* (Tulving, 1989). For example, your memory of the rules of tennis is a semantic memory; your memory of the most recent time you played tennis is an episodic memory. Your memory of the periodic table of chemistry is a semantic memory; your recollection of breaking a test tube in a chemistry lab is an episodic memory.

Episodic memories are in many cases more fragile than semantic memories. For example, people sometimes remember a statement they have heard (a semantic memory) but forget when, where, and from whom they heard it (an episodic memory). That phenomenon is known as **source amnesia,** *remembering the content but not the context of learning it.* Because of source amnesia, people sometimes confuse reliable information with the unreliable: "Did I hear this idea from my professor or was it on *South Park*? Did I read about brain transplants in *Scientific American* or in the *National Enquirer*?" As a result, you might dismiss an idea at first ("Oh, that's just a rumor!" or "Oh, that's just my nutty roommate's idea!") but later remember the idea, forget where you heard it, and start to take it seriously (Johnson, Hashtroudi, & Lindsay, 1993; Riccio, 1994).

CONCEPT CHECK

6. Is your memory of your current mailing address a semantic memory or an episodic memory? What about your memories of the day you moved to your current address? (Check your answers on page 241.)

THE MESSAGE
Varieties of Memory

Although researchers still disagree with one another on many points about memory, they do agree about what memory is *not*: Memory is not a single store into which we simply dump things and later take them out. When Ebbinghaus conducted his studies of memory in the late 1800s, he thought he was measuring the properties of memory, period. We now know that the properties of memory depend on the type of material being memorized, the individual's experience with similar materials, the method of testing, and the recency of the event. Memory is not one process, but many.

SUMMARY

* *Ebbinghaus's approach.* Hermann Ebbinghaus pioneered the experimental study of memory by testing his own ability to memorize and retain lists of nonsense syllables. Although the general principles he reported were valid, his measurements of the speed of learning and forgetting do not apply to all memory. (page 229)
* *Variations in memory strength.* People remember material best if they have minimal interference from similar materials, if the material is meaningful and distinctive, and if the method of testing is capable of detecting a weak memory. (page 230)
* *The information-processing model.* Psychologists distinguish among various types of memory. According to the information-processing model of memory, information is stored first as short-term memories and later processed to become long-term memories. (page 233)
* *Memory capacity.* Short-term memory has a capacity of only about seven items in normal adults, although chunking can enable us to store much information in each item. Long-term memory has a very large, not easily measured capacity. (page 234)

* *Weaknesses of the traditional information-processing model.* Contrary to what psychologists once believed, how long an item remains in short-term memory is a poor predictor of whether it will enter long-term memory. Furthermore, some people with a defective short-term memory have a normal long-term memory. Evidently, short-term memory is not simply a holding station on the way to long-term memory. (page 237)
* *Working memory.* As an alternative to the traditional description of short-term memory, most current researchers identify working memory as a system for storing and processing several kinds of current information. (page 238)
* *Semantic and episodic memories.* An additional distinction can be drawn between semantic memories (memories of general principles) and episodic memories (memories of personal experiences). In most cases episodic memories are more fragile and more easily lost or distorted. (page 239)

Suggestion for Further Reading

Schacter, D. L. (1996). *Searching for memory.* New York: Basic Books. Discusses current theories and research in an accessible manner.

Terms

proactive interference the hindrance that an older memory produces on a newer one (page 231)
retroactive interference the impairment that a newer memory produces on an older one (page 231)
von Restorff effect the tendency to remember distinctive or unusual items on a list better than other items (page 232)
recall a method of testing memory by asking someone to produce a certain item (such as a word) (page 232)
cued recall a method of testing memory by asking someone to remember a certain item after being given a hint (page 232)
recognition a method of testing memory by asking someone to choose the correct item from a set of alternatives (page 232)
savings method or **relearning method** a method of testing memory by measuring how much faster someone can relearn something than learn something for the first time (page 233)
information-processing model the view that information is processed, coded, and stored in various ways in human memory as it is in a computer (page 233)
sensory store a very brief storage of sensory information (page 234)
short-term memory a temporary storage of a limited amount of information (page 234)
long-term memory a relatively permanent store of information (page 234)
chunking the process of grouping digits or letters into meaningful sequences (page 235)
consolidation the formation and strengthening of long-term memories (page 237)
working memory a system that processes and works with current information, including three components—a central executive, a phonological loop, and a visuospatial sketchpad (page 238)
semantic memory memory of general principles (page 239)
episodic memory memory for specific events in a person's life (page 239)
source amnesia remembering content but not the context of learning it (page 239)

Answers to Concept Checks

1. It is due to proactive interference—interference from memories learned earlier. Much of the difficulty of retrieving long-term memories is due to proactive interference. (page 231)
2. First, the subject learns the response; second, the subject learns the extinction of the response. If the first learning proactively interferes with the later learning, spontaneous recovery will result. (page 231)
3. **a.** savings; **b.** recall; **c.** recognition; **d.** cued recall. (page 233)
4. Short-term memory has a limited capacity of about seven items in normal adults, whereas long-term memory has an extremely large capacity (difficult to estimate). Both short-term memory and long-term memories are more rapidly forgotten in the presence of interference. (page 237)
5. Readers with a shorter phonological loop should show greater comprehension of Italian, which will seldom require them to remember a word for very long before deciding on its meaning. (page 239)
6. Your memory of your current address is a semantic memory. Your memory of the events of moving day is an episodic memory. (page 239)

Answer to Other Questions in the Text

A. Hermann Melville, Jane Austen, Agatha Christie, Arthur Conan Doyle, Maya Angelou, Leo Tolstoy, James Kalat, Geoffrey Chaucer, Charles Darwin, Margaret Mitchell, Alice Walker, Victor Hugo. (page 233)

Web Resources

The Magic Seven, Plus or Minus Two
www.well.com/user/smalin/miller.html
This is George Miller's classic article about the limits of short-term memory, complete with graphs and references, as it originally appeared in the *Psychological Review* in 1956.

MODULE 7.2

Memory Improvement

How can we improve our memory?

There you are, sitting in class taking a geography test, unable to remember the major rivers of Africa. You remember reviewing that section in your book last night; you even remember that it was on the upper left side of the page, and there was a diagram to the right of it. You even remember the cappuccino you were sipping as you were studying. And that it was about 9:30 P.M. at the time. You just don't remember the names of the rivers.

How is it that we sometimes remember so much useless information while forgetting what we really wanted to remember? And how can we improve our memory?

The short answer is that, to improve your memory, you must improve the way you store the material in the first place. The rest of this module elaborates on that point.

The Influence of Emotional Arousal

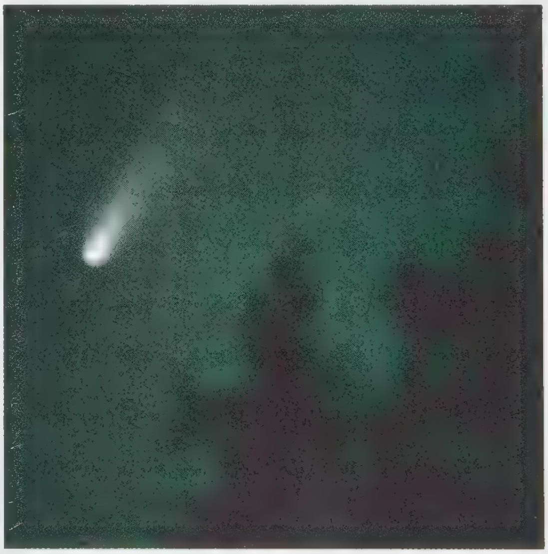
People usually remember emotionally arousing events. Chances are you can vividly remember your first day of college, your first kiss, the time your team won the big game, and various times when you were extremely frightened or extremely excited.

The effects of arousal on memory have been known for centuries. In England in the early 1600s, when people sold land, they did not yet have the custom of recording the sale on paper. (Paper was expensive and most people were illiterate anyway.) Instead, residents of the city or village would gather together in a ceremony; someone announced the sale and instructed everyone to remember it. The children's memory was especially important, because they would live the longest. To increase the chances that the children would remember, the adults would kick the children while telling them about the business deal. (Avoiding such abuse is just another of the many benefits of literacy.)

Unfortunately, emotional intensity does increase the vividness and intensity of a memory, but it does not guarantee the memory's accuracy. Many people report "flashbulb" memories of where they were, what they were doing, or even the weather at the time they heard shocking news—the assassination of John Kennedy, the explosion of the *Challenger,* the death of Princess Diana, and so on. One investigator asked English people to remember the moment they first heard about the Hillsborough football disaster, when a stampede of fans trying to enter the stadium crushed 95 people to death. Large numbers of people claimed to have clear recollections of the moment when they first heard the news. However, when the same people were interviewed on different occasions, the "clear, vivid" memories they once reported were not always the same as what they reported another time (Wright, 1993). So, emotionally charged memories are subject to distortion, just as other memories are.

Why are emotionally arousing events so memorable (even if the memories are not always accurate)? First, emotionally exciting events stimulate certain kinds of norepinephrine synapses in the brain, known as β-adrenergic synapses, which enhance memory storage. Drugs that block those synapses decrease the storage of emotionally exciting events (Cahill, Prins, Weber, & McGaugh, 1994). Second, exciting experiences arouse your sympathetic nervous system, increasing the conversion of stored glycogen into glucose (a sugar) and therefore raising the level of blood glucose (McGaugh, 1990). Recall from Chapter 3 that glucose is the brain's primary fuel; elevating the glucose level facilitates brain functioning.

Could you enhance your memory by taking drugs that stimulate β-adrenergic synapses? In principle, maybe;



Do you remember the first time you saw a comet? Most people recall emotionally arousing events, sometimes in great detail, although not always accurately.

however, such drugs produce unwanted side effects, including headaches, nausea, and heart problems. Could you enhance your memory by increasing your blood glucose levels? Yes, but don't try to do that by eating chocolate. Excess carbohydrates raise your blood glucose only briefly before you store them, so your waistline will increase more than your test scores. The best solution is to try to increase your interest in the material and, therefore, your emotional response to it.

TABLE 7.3 Levels-of-Processing Model of Memory

Superficial processing	*Simply repeat the material to be remembered: "Hawk, Oriole, Tiger, Timberwolf, Blue Jay, Bull."*
Deeper processing	*Think about each item. Note that two start with T and two with B.*
Still deeper processing	*Note that three are birds and three are mammals. Also, three are major league baseball teams and three are NBA basketball teams. Use whichever associations mean the most to you.*

Meaningful Storage and Levels of Processing

If you want to memorize something, such as a definition, you might try repeating it many times. Other things being equal, repetition does aid memory, but other things are seldom equal. To illustrate: Examine Figure 7.15, which shows a real U.S. penny and 14 fakes. If you live in the United States, you have seen pennies countless times, but can you now identify the real one? Most U.S. citizens cannot (Nickerson & Adams, 1979). (If you do not have a penny in your pocket, check answer B on page 248. If you are not from the United States, try drawing the front and back of a common coin in your own country.) In short, mere repetition, such as looking at a coin many times, does not guarantee a strong memory.

TRY IT YOURSELF

According to the **levels-of-processing principle** (Craik & Lockhart, 1972), *how easily we can retrieve a memory depends on the number and types of associations we form.* When you read something—this chapter, for example—you might simply read over the words, giving them no more thought than necessary to complete your reading assignment. In that case, you have engaged in shallow processing, and whatever memories you store will be hard to retrieve at test time. Alternatively, you might stop and think about various points that you read, relate them to your own experiences, and think of your own examples of the principles discussed. The more ways you think about the material, the deeper your processing is and the more easily you will remember it later. Table 7.3 summarizes this model.

As an example, imagine several groups of students who study a list of words in several ways. One group simply reads the list over and over, and a second counts the letters in each word. Both procedures produce *superficial processing* and poor recall later. A third group tries to think of a synonym for each word or tries to use each word in a sentence. As the students think about the words, they store them at a *deeper level of processing,* so they will recall them better than the first two groups. Students in the fourth group try to relate each word to themselves: "How does this apply to me? What experiences have I had that relate to this word?" This group does even better than the third group. For a while, psychologists thought that relating words to yourself produces a special kind of strengthening, but later research found equally strong memories in students who tried to relate each word to their mothers (Symons & Johnson,

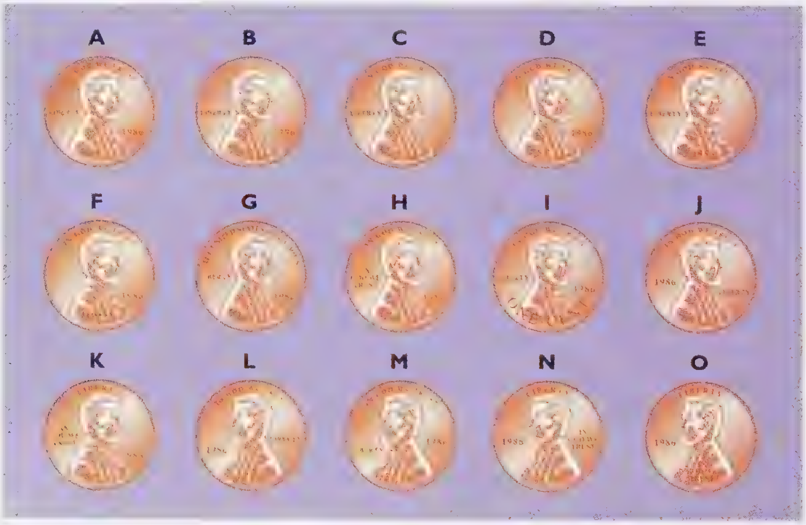


FIGURE 7.15 Can you spot the genuine penny among 14 fakes? (Based on Nickerson & Adams, 1979.) If you're not sure (and you don't have a penny with you), check answer B on page 248.

Most actors prepare for a play by spending much time thinking about the meaning of what they say (a deep level of processing) and spending only a little time simply repeating the words (a shallow level of processing).

1997). The conclusion is that memory grows stronger when people elaborate and organize the material and relate it to other things that they know and care about.

You can improve your level of processing in one of two ways (Einstein & Hunt, 1980; McDaniel, Einstein, & Lollis, 1988): First, you can think about each individual item on the list you are trying to remember. Second, you can look for relationships among the items. You might notice, for example, that the list consists of five animals, six foods, four methods of transportation, and five objects made of wood. One of the skills we gain through education is learning to organize material in this way. According to one study of Kpelle children in West Africa, teenage children who had gone to school organized a list of words into categories, such as foods, types of clothing, and hunting materials. Unschooled children did not sort the list into categories and generally did not remember the words very well (Scribner, 1974).

CONCEPT CHECKS

7. Many of the best students in a course (those who get the best grades) read the assigned text chapters more slowly than average. Why?
8. Most actors and public speakers who have to memorize lengthy passages spend little time simply repeating the words and more time thinking about them. Why? (Check your answers on page 248.)

Self-Monitoring of Understanding

Whenever you are studying a text, you must periodically decide, "Should I keep on studying this section, or do I already understand it well enough?" Even when you are reading a single sentence, you have to decide whether you understand the sentence or whether you should stop and reread it. This sentence was once published in the student newspaper at North Carolina State University:

> He said Harris told him she and Brothers told French that grades had been changed.

Ordinarily, when good readers encounter such a confusing sentence, they notice their own confusion and reread the sentence or, if necessary, the whole paragraph. Poor readers tend to read at the same speed for both easy and difficult materials, instead of slowing down when they come to difficult sentences.

Monitoring one's own understanding is difficult and often inaccurate, yet it is not impossible. After you have read a chapter, you can probably make a somewhat accurate (better than chance) estimate of how well you will perform when you are tested on the material later (Maki & Serra, 1992). If you wait a while (at least a few minutes) after studying something and then try to predict how well you will remember it later, your prediction will improve (Weaver & Kelemen, 1997). Evidently, you monitor how much you have already forgotten and use that to estimate your later memory losses.

When Mozart was a boy he visited the Vatican, where he heard a performance of a piece of music. The next day, he had a handwritten score of the piece. The Pope was furious, because he had decreed that no one could copy the score of that music, but Mozart had written down the entire piece from memory—eight voice parts—after one hearing. The Pope was so impressed, he awarded Mozart a medal.

People need to monitor their understanding of a text to decide whether to continue studying or whether they already understand it well enough. Most readers have trouble making that judgment correctly.

One systematic way to monitor your own understanding of a text is by using the **SPAR method:**

- S*urvey.* Get an overview of what the passage is about. Scan through it; look at the boldface headings; try to understand the organization or goals of the passage.
- P*rocess meaningfully.* Read the material carefully. Think about how you could use the ideas, or how they relate to other things you have read. Evaluate the strengths and weaknesses of the argument. The more actively you think about what you read, the better you will remember the material.
- A*sk questions.* If the text provides questions, like the concept checks in this text, try to answer them. Then pretend you are the instructor; write the questions you would ask on a test, and then answer them yourself. In the process, you will discover which sections of the passage you need to reread.
- R*eview.* Wait a day or so, and then retest your knowledge. Spacing out your study over time increases your ability to remember it over the long term.

The Effects of Spacing Out Study Sessions

Suppose someone reads you a list of 20 words and asks you to recall as many of them as possible. You cannot store that many words in your phonological loop, but you will probably remember a few words, *especially the items at the beginning and end of the list.*

That tendency, known as the **serial-order effect,** includes two aspects: The *primacy effect* is the tendency to remember the first items, and the *recency effect* is the tendency to remember the last items. One explanation for the primacy effect is that the listener has a chance to rehearse the first few items for a few moments by themselves with no interference from the others. One explanation for the recency effect is that the last items are still in the listener's phonological loop at the time of the test.

The phonological loop cannot provide the whole explanation for the recency effect, however. In one study, British rugby players were asked to name the teams they had played against in the current season. Players were most likely to remember the last couple of teams they had played against, thus showing a clear recency effect, even though they were recalling events that occurred weeks apart (Baddeley & Hitch, 1977). (The phonological loop holds information for only seconds.)

So, studying material—or, rather, *reviewing* material—shortly before a test is likely to improve recall, via the recency effect. Now consider something you once studied and have not reviewed since then, such as a foreign language you studied several years ago. If you resumed your study, would much of it come back to you?

Harry Bahrick (1984) tested people who had studied Spanish in school anywhere from 1 to 50 years previously. People who had studied it 1 or 2 years ago remembered more than those who had studied it 3 to 6 years ago, but beyond 6 years the retention appeared to be stable (Figure 7.16). In other words, we do not completely forget old memories even if we seldom use them.

In a later study, Bahrick and members of his family studied foreign-language vocabulary either on a moderately frequent basis (practicing once every 2 weeks) or on a less-frequent basis (as seldom as once every 8 weeks); both schedules eventually reached the same total study time.

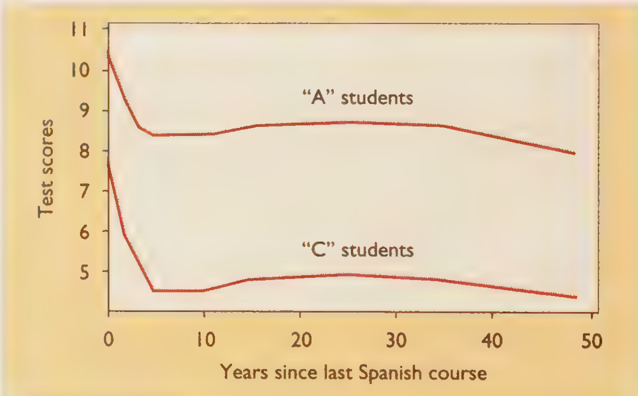


FIGURE 7.16 (Left) Spanish vocabulary as measured by a recognition test shows a rapid decline in the first few years but then long-term stability. (From Bahrick, 1984.)

The result: More frequent study led to faster learning; however, less frequent study led to better long-term retention, as measured years later (Bahrick, Bahrick, Bahrick, & Bahrick, 1993).

The principle here is a general one: *If you want to remember something for as long as possible*—and presumably that is why you are getting an education in the first place—you should study and review under varying conditions with substantial intervals between study sessions. Spreading out your study will slow down your original learning but will improve your ability to recall the information long afterward (Schmidt & Bjork, 1992).

CONCEPT CHECKS

9. The results shown in Figure 7.16 demonstrate how well people recognize the Spanish they studied years ago. Might they actually remember more than this figure indicates? If so, how could you determine how much more they remember?

10. If you want to do well on the final exam in this course, what should you do now—review this chapter or review the first three chapters in the book? (Check your answers on page 248.)

FIGURE 7.17 According to the principle of encoding specificity, the way we code a word during original learning determines which cues will remind us of that word later. For example, when you hear the word *queen*, you may think of that word in any of several ways. If you think of *queen bee*, then the cue *playing card* will not remind you of it later. If you think of the *Queen of England*, then *chess piece* will not be a good reminder.

The Use of Special Coding Strategies

A librarian who places a new book on the shelf also enters some information into the retrieval system, so that anyone who knows either the title, author, or topic can find the book. Similarly, when you store a memory, you store it in terms of **retrieval cues,** *associated information that might help you regain the memory later.* We shall examine two illustrations of retrieval cues, the encoding specificity effect and the use of mnemonic devices.

Encoding Specificity

When you learn something, the associations you form with the learned material become retrieval cues that can remind you of the material later. According to the **encoding specificity principle** (Tulving & Thomson, 1973), *the associations you form at the time of learning will be the most effective retrieval cues* (Figure 7.17).

Here is an example (modified from Thieman, 1984).

TRY IT YOURSELF

First, read the pairs of words (in psychological jargon, *paired associates*) in Table 7.4a. Then, turn to Table 7.4b on page 247. For each of the words on that list, try to recall a related word on the list you just read. *Do this now.* (The answers are on page 248, answer C.)

Most people find this task difficult. Because they initially coded the word *cardinal* as a type of clergyman, for example, they do not think of it when they see the retrieval cue *bird.* Consequently, you can improve your memory by storing information in terms of the same retrieval cues you will most likely use when you try to recall it. (Or store it with as many different retrieval cues as possible.)

The principle of encoding specificity extends to practically any aspect of experience at the time of storage. If you experience something while you are in a particular mood, you are somewhat more likely to think of it again when you are in the same mood (Bower, 1994). If you learn something while under the influence of a drug, such as a tranquilizer, you will probably remember it more easily when you take that drug again than at other times. **State-dependent memory** is *the tendency to remember something better if your body is in the same condition during recall as it was*

TABLE 7.4a

Clergyman — Cardinal	
Trinket — Charm	
Social event — Ball	
Shrubbery — Bush	
Inches — Feet	
Take a test — Pass	
Weather—Fair	
Geometry — Plane	
Tennis — Racket	
Stone — Rock	
Magic — Spell	
Envelope — Seal	
Cashiers — Checkers	

Nobel Peace Prize Winners

1901	H. Dunant and F. Passy
1902	E. Ducommun and A. Gobat
1903	Sir W. R. Cremer
1904	Institute of International Law
1905	Baroness von Suttner
1906	T. Roosevelt
1907	E. T. Moneta and L. Renault
1908	K. P. Arnoldson and F. Bajer
1909	A. M. F. Beernaert and Baron d'Estournelles de Constant
1910	International Peace Bureau
1911	T. M. C. Asser and A. H. Fried
1957	L. B. Pearson
1958	G. Pire
1959	P. J. Noel-Baker
1960	A. Luthuli
1961	D. Hammarskjöld (posthumously)
1962	L. Pauling (awarded 1963)
1990	M. Gorbachev
1991	A. S. Suu Kyi
1992	Rigoberta Menchú
1993	Nelson Mandela and Frederik W. de Klerk
1994	Yasir Arafat, Yitzhak Rabin, and Shimon Peres
1995	Joseph Rotblat and Pugwash Conferences on Science and World Affairs
1996	Carlos Felipe Ximenes Belo and José Ramos-Horta
1997	Jody Williams and International Committee to Ban Landmines

FIGURE 7.18 A list of Nobel Peace Prize winners: Mnemonic devices can be useful when people try to memorize long lists like this one.

during the original learning. The point is that, if you want to remember something easily and under a wide variety of conditions, study it under a wide variety of conditions, not just one.

Mnemonic Devices

If you needed to memorize something lengthy and not terribly exciting—for example, a list of all the bones in the body—how would you do it? One effective strategy is to attach systematic retrieval cues to each term, so that you can remind yourself of the terms when you need them.

A **mnemonic device** is *any memory aid that is based on encoding each item in a special way.* The word *mnemonic* (nee-MAHN-ik) comes from a Greek root meaning "memory." (The same root appears in the word *amnesia,* "lack of memory.")

Mnemonic devices come in many varieties. Some are simple: thinking up a little story that reminds you of each item to be remembered (such as "Every Good Boy Does Fine" to remember the notes EGBDF on the musical staff). Suppose you had to memorize the list of Nobel Peace Prize winners (Figure 7.18). You might try making up a little story: "Dun (Dunant) passed (Passy) the Duke (Ducommun) of Gob (Gobat) some cream (Cremer). That made him internally ILL (Institute of International Law). He suited (von Suttner) up with some roses (Roosevelt) and spent some money (Moneta) on a Renault (Renault) . . . ." You'd still have to study the names, but your story might help.

Another effective mnemonic device is the **method of loci** (method of places). *First, you memorize a series of places, and then you use a vivid image to associate each of these locations with something you want to remember.* For example, you might start by memorizing every location along the route from your dormitory room to, say, your psychology classroom. Then you link the locations, in order, to the names.

Suppose the first three locations you pass are the desk in your room, the door to your room, and the corridor. You should first form a mental image linking the first pair of Nobel Peace Prize winners, Dunant and Passy, to the first location, your desk. You might imagine a Monopoly game board on your desk, with a big sign "DO NOT (Dunant) PASS (Passy) GO." Then you link the second pair of names to the second location, your door: A DUKE student (as in Ducommun) is standing at the door, giving confusing signals. He says "DO COME IN (Ducommun)" and "GO BACK (Gobat)." Then you link the corridor to Cremer, perhaps by imagining someone has spilled CREAM (Cremer) all over the floor (Figure 7.19). You continue in this manner until you have linked every name to a location. Now, if you can remember all those locations in order and if you have good visual images for each one, you will be able to recite the list of Nobel Peace Prize winners.

A similar mnemonic device is called the **peg method.** *You start by memorizing a list of objects, such as "One is a bun, two is a shoe, three is a tree, . . ." Then you form mental images to link the names with these peg words,* just as you would with the method of loci. For example, for number one, "I ate a BUN at the DUNE PASS (Dunant, Passy)," imagining *dune pass* as a passageway between sand

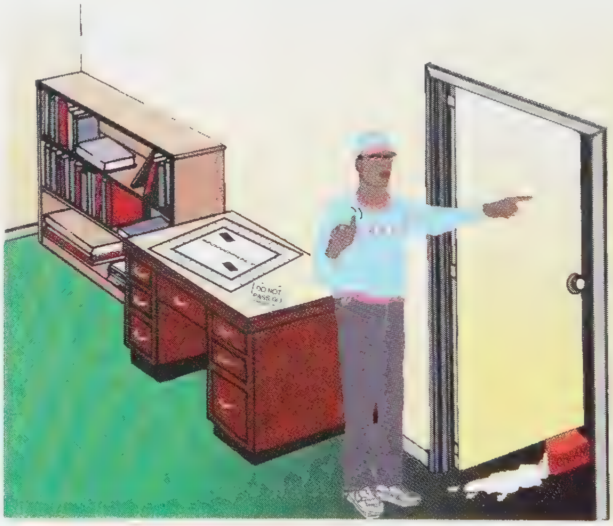


FIGURE 7.19 The method of loci is one of the oldest mnemonic devices. First learn a list of places, such as "my desk, the door of my room, the corridor, . . ." Then link each of these places to the items on a list of words or names, such as a list of the names of Nobel Peace Prize winners.

TABLE 7.4b

Instructions: For each of these words, write a related word that you remember from the *second column* of the list in Table 7.3.

Exhibition —
Animal —
Part of body —
Transportation —
Football —
Crime —
Former U.S. president —
Music —
Personality —
Write —
Bird —
Board game —
Sports —

dunes. Later, you use all your peg words to help you remember the list of names. To use mnemonic devices well, you need to be clever and resourceful in producing images. (What image can you devise for Nobel Peace Prize winner Mikhail Gorbachev?)

How useful are elaborate mnemonic devices? Learning to use mnemonics takes some time and effort, but the rewards are substantial for someone who wants to memorize long lists of names or words. Some people find mnemonics useful for remembering people's names. (For example, you might remember someone named Harry Moore by picturing him as "more hairy" than everyone else.) Few people find mnemonics useful for everyday tasks such as remembering where you parked your car, but mnemonics can be helpful in their proper place.

THE MESSAGE
Improving One's Memory

We often speak of "storing" and "retrieving" memories, as if we were putting items on a shelf and then taking them off. For some purposes, this analogy is satisfactory, but only if we don't take it too seriously. The more you know about something and the more interested you get, the easier it is to establish new memories and find them when you need them. It is as though putting a certain kind of item on your "shelf" makes it easier to add more items of the same kind.

SUMMARY

* *Emotional arousal.* Emotionally exciting events tend to be remembered vividly, if not always accurately. Such events stimulate β-adrenergic synapses in the brain and act to increase blood glucose levels. (page 241)

* *Levels-of-processing principle.* According to the levels-of-processing principle, a memory becomes stronger (and easier to recall) if we think about the meaning of the material and relate it to other material. (page 242)

* *Self-monitoring of understanding.* Good readers check their own understanding of a text and slow down or reread when they do not understand. Self-monitoring is difficult, but people can learn techniques to improve their self-monitoring. (page 243)

* *Timing of study.* Studying under consistent conditions as close as possible to the time of the test will increase your memory during the test. However, reviewing under varied conditions at long, irregular intervals is better for establishing memories that will be available at varied times and places. (page 244)

* *Encoding specificity.* When we form a memory, we store it with links to the way we thought about it at that time. When we try to recall the memory, a cue is most effective if it is similar to the links we formed at the time of storage. (page 245)

* *Mnemonics.* Specialized techniques for establishing systematic retrieval cues can help people to remember ordered lists of names or terms. (page 246)

Suggestion for Further Reading

Cermak, L. S. (1975). *Improving your memory.* New York: McGraw-Hill. A lively book about mnemonic devices and other ways to improve memory.

Terms

levels-of-processing principle the concept that the number and types of associations established during learning determines the ease of later retrieval of a memory (page 242)

SPAR method a systematic way to monitor and improve understanding of a text by surveying, processing meaningfully, asking questions, and reviewing (page 244)

serial-order effect the tendency to remember the items near the beginning and end of a list better than those in the middle (page 244)

retrieval cue information associated with remembered material, which can be useful for helping to recall that material (page 245)

encoding specificity principle the tendency for the associations formed at the time of learning to be more effective retrieval cues than other associations (page 245)

state-dependent memory the tendency to remember something better if your body is in the same condition during recall as it was during the original learning (page 245)

mnemonic device any memory aid that is based on encoding each item in a special way (page 246)

method of loci a mnemonic device that calls for linking the items on a list with a memorized list of places (page 246)

peg method a mnemonic device in which a person first memorizes a list of objects and then forms mental images linking those objects ("peg words") to a list of names to be memorized (page 246)

Answers to Concept Checks

7. Students who read slowly and frequently pause to think about the meaning of the material are engaging in deep processing and are likely to remember the material well, probably better than those who read through the material quickly. (page 243)
8. Simply repeating the words would produce a shallow level of processing and poor retention. A more efficient means of memorizing is to spend most of one's time thinking about the meaning of the speech and only a little time actually memorizing the words. (page 243)
9. People would probably show even greater retention if they were tested by the savings method. (page 245)
10. To prepare well for the final exam, you should review all the material at irregular intervals. Thus, you might profit by skimming over Chapters 2 and 3 right now. Of course, if you have a test on Chapter 7 in a day or two, your goal is different and your strategy should be different. (page 245)

Answers to Other Questions in the Text

B. The correct coin is A. (page 242)
C. (page 247)

Exhibition—Fair
Animal—Seal
Part of body—Feet
Transportation—Plane
Football—Pass
Crime—Racket
Former U.S. president—Bush
Music—Rock
Personality—Charm
Write—Spell
Bird—Cardinal
Board game—Checkers
Sports—Ball

Web Resources

Memory Techniques and Mnemonics
www.demon.co.uk/mindtool/memory.html
MindTools offers commercial software designed to improve memory, and more than 25 free articles about memory and how to improve it, stress management, time management, and other topics.

MODULE 7.3

Memory Loss

What are some of the causes of memory loss?

What do various types of memory loss teach us about memory?

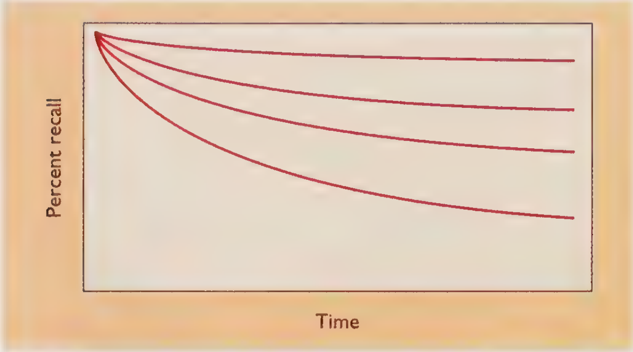
Computer operators are advised to protect their computers from strong magnetic fields. Suppose you defied that advice and passed your computer through the strongest magnetic field you could find. Chances are, you would erase all your computer's memory, but suppose it erased all of the text files but none of the graphics files. Or perhaps it left all the old memories intact but made it impossible to store new ones. From the damage, you would receive hints about how your computer's memory works.

The same is true of human memory. Passing your head by a magnet would not erase your memories. (At least not with ordinary magnets; I don't know what would happen at extreme levels.) However, the kinds of damage that do impair memory affect different kinds of memory in different ways. The study of severe memory loss helps us to understand systems of memory.

"Normal" Forgetting

TRY IT YOURSELF

Pick your favorite meal. I'm going to imagine that you said pizza, but substitute something else if you wish. List all the pizza meals you can remember. You will probably start with your most recent pizza meals and then add some especially interesting or unusual meals from long ago. You will recall more recent than remote examples, but you will recall a few from many years ago, and if you still recall a memory that is a few years old now, you will probably continue to remember it indefinitely. In short, most memories fade rapidly, but the longer you have held onto a memory, the less it tends to fade. The overall trend of memory and forgetting looks approximately as shown in the following diagram (Rubin & Wenzel, 1996). The family of curves shows that the rate of forgetting varies among individuals and even among situations for a given individual. Nevertheless, the shape of the curve is consistent.



Why do we forget? Let us review the possibilities already discussed in this chapter. One explanation is interference. For example, sports fans will forget most of the details of a game they saw a year ago, largely because of interference from their memories of all the games they saw before it (proactive interference) and after it (retroactive interference).

A second explanation is that memory traces gradually decay, like old photographs that fade in the sunlight. The effects of decay and interference are difficult to disentangle experimentally. That is, during any delay, a memory is subject to the effects of both time and interference from other experiences. Nevertheless, researchers have considered decay a likely explanation of forgetting from working memory, because some information in the phonological loop disappears within seconds if it is not rehearsed. Decay is a less likely explanation of forgetting from long-term memory, because some long-term memories remain available after years even without use.

A third possible explanation is that memories seem to be forgotten because we no longer have the appropriate retrieval cues. For example, if you learned certain facts in an eighth-grade geography class, the retrieval cues included the context—when and where you learned it (the classroom, the city, the other students around you). If you try to remember these facts years later, you no longer have the contextual cues to help remind you. Sometimes, if people return to the place where they learned something (at least in their imagination), they remember information that they thought they had forgotten.

You are, of course, most likely to forget material when you were paying little attention at the time you learned it. In particular, people often remember an idea and forget where they heard it. Have you ever told someone something—a joke, a bit of gossip perhaps—only to have that person say, "Yes, I was the one who told *you* that"? In one experiment, students sat in a group trying to think of solutions to a problem. A week later, each student was asked separately to think of other possible solutions. Many of them offered, as their own "original" idea, something that they had heard another student suggest the previous week (Marsh, Landau, & Hicks, 1997). They had simply forgotten where they got the

Bob Williams was one of 40 aging former paratroopers who reenacted his parachute jump on the 50th anniversary of D day. Ordinarily we forget most events from 50 years ago. However, this was a very distinctive, emotionally arousing event, followed by very few other experiences that were similar enough to produce interference. Returning to the scene of the original jump provides contextual cues that would help to retrieve the memory.

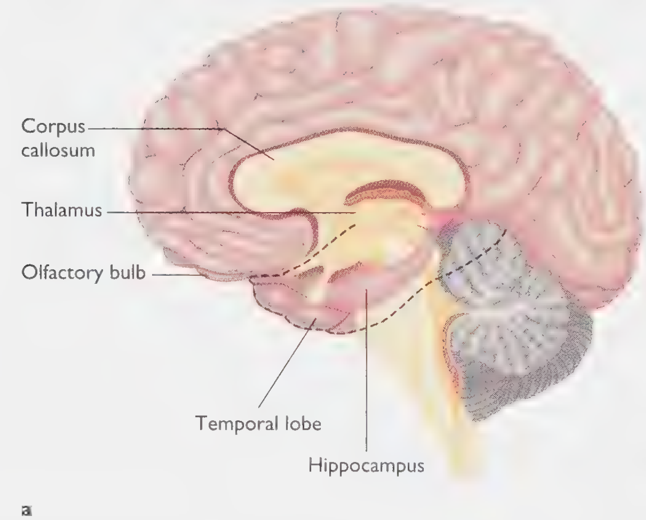
idea. The result is unintentional plagiarism: More often than we realize, we claim an idea to be new, not remembering that we first heard it from someone else.

Amnesia After Brain Damage

In contrast to normal forgetting, **amnesia** is a *severe loss or deterioration of memory.* Even in the most severe cases of amnesia, people do not forget everything they have ever learned. The specific deficits of amnesic patients tell us much about the distinctions among the various types of memory. We shall consider amnesia based on several types of brain damage, plus amnesia of infancy and old age.

The Case of H. M., a Man with Hippocampal Damage

In 1953, a man with the initials H. M. was subjected to unusual brain surgery in an attempt to control his severe epilepsy. Even while taking antiepileptic drugs, H. M. had suffered such frequent and severe seizures that he was unable to keep a job or live a normal life. In desperation, surgeons removed his **hippocampus,** *a large forebrain structure in the interior of the temporal lobe* (Figure 7.20),



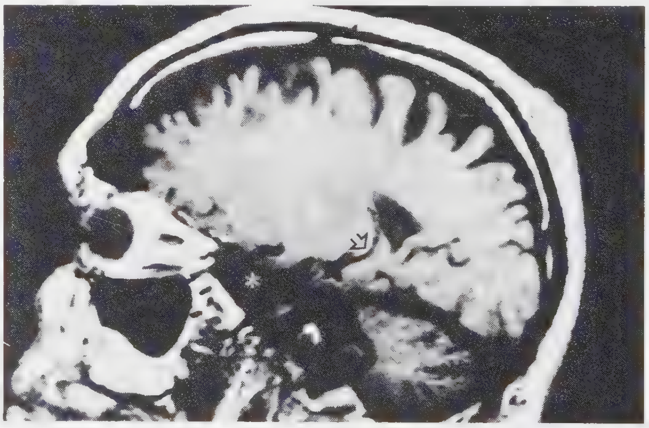


FIGURE 7.20 The hippocampus is a large subcortical structure of the brain. After damage to the hippocampus and related structures, patient H. M. had great difficulty storing new long-term memories. The photo shows a scan of the brain of H. M. formed by magnetic resonance imaging. The asterisk indicates the area from which the hippocampus is missing. The arrow indicates a portion of the hippocampus that is preserved. (Photo courtesy of Suzanne Corkin and David Amaral.)

where they believed his epileptic seizures were originating. They also removed some surrounding brain areas. At the time, researchers knew little about the functions of the hippocampus or the kinds of memory deficits likely to result from its removal.

The results of the surgery were favorable in some regards. H. M.'s epileptic seizures decreased in frequency and severity. His personality and intellect remained the same; in fact, his IQ score increased slightly after the operation, presumably because of the decreased epileptic interference.

However, he suffered severe memory problems (Corkin, 1984; Milner, 1959), particularly a massive **anterograde** (ANT-eh-ro-grade) **amnesia,** *inability to store new long-term memories.* For years after the operation, he cited the year as 1953 and his own age as 27. Later, he took wild guesses (Corkin, 1984). He would read the same issue of a magazine repeatedly without realizing that he had read it before. He could not even remember where he had lived for the last few years. He also suffered a moderate **retrograde amnesia,** *loss of memory for events that occurred shortly before the brain damage;* see Figure 7.21). That is, he had some trouble recalling events that had happened within the last 1 to 3 years before the operation, although he could recall older events. People who suffer a head injury with loss of consciousness generally have retrograde amnesia for the events leading up to the injury.

H. M. could form normal short-term memories, such as repeating a brief list of items, and if he was permitted to rehearse them without distraction, he could even retain the list of items for several minutes. He could also learn new skills, as we shall see in a moment.

Like Rip van Winkle (the story character who slept for 20 years and awakened to a vastly changed world), H. M. became more and more out of date with each passing year (Gabrieli, Cohen, & Corkin, 1988; Smith, 1988). He did not recognize the names or faces of people who became famous after the mid-1950s. He could not name the president of the United States even when Ronald Reagan was president, though he remembered Reagan as an actor from before 1953. He did not understand the meaning of words that had entered the English language after the time of his surgery. For example, he guessed that *biodegradable* means "two grades," that *soul food* means "forgiveness," and that a *closet queen* is a "moth." For H. M., watching the evening news was like visiting another planet.

In spite of H. M.'s massive memory difficulties, he can still acquire new skills and retain them later. We refer to *skill retention* as **procedural memory,** in contrast to **declarative memory,** *the ability to recall factual information.* For example, H. M. has learned to read material written in mirror fashion (Cohen & Squire, 1980):

with the words reversed like this

Although H. M. has learned to read mirror writing, he does not remember having learned it. He has also learned a simple finger maze, and he has learned the correct solution to the Tower of Hanoi puzzle shown in Figure 7.22 (Cohen, Eichenbaum, Deacedo, & Corkin, 1985). He does not *remember* learning these skills, however. He claims that he has never seen any of these tasks before, and he is always a bit surprised by his success.

Because H. M.'s memory deficits are so severe, surgeons today would not attempt the same surgery. Occasionally, people suffer accidental damage to the hippocampus from stroke or other means; those people also show extensive memory impairment, especially when describing events in their lives (Reed & Squire, 1997; Vargha-Khadem et al., 1997). Both human and animal studies confirm that memories are not stored *in* the hippocampus, but the hippocampus must function in order to store certain kinds of memories. Once these memories are stored, however, they can persist after removal of the hippocampus. (After all, H. M. remembers most of what happened before his operation.)

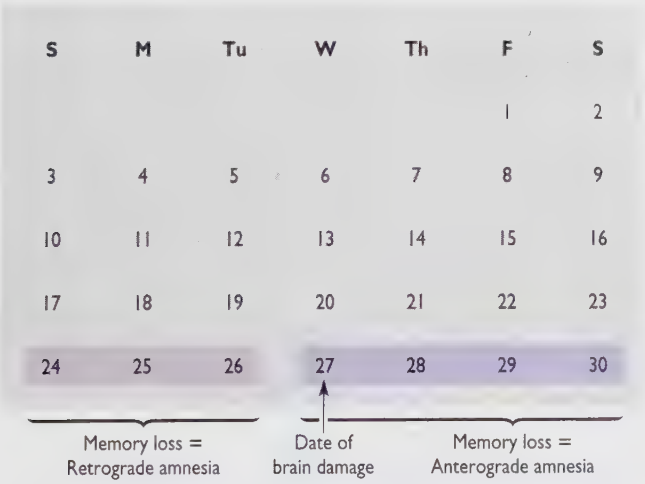


FIGURE 7.21 Retrograde amnesia is loss of memory for events in a certain period *before* brain damage or another trauma. Anterograde amnesia is difficulty forming new memories *after* some trauma.

Frontal-Lobe Amnesia

Amnesia can also arise after damage to the frontal lobes, especially the prefrontal cortex (see Figure 7.20). Because the frontal lobes receive a great deal of input from the hippocampus, the symptoms of frontal-lobe damage overlap those of hippocampal damage. However, frontal-lobe damage produces some special memory impairments of its own.

Frontal-lobe damage can be either the result of a stroke or of trauma to the head. Frontal-lobe deterioration is also commonly associated with **Korsakoff's syndrome,** *a condition caused by a prolonged deficiency of vitamin* B_1*, usually as a result of chronic alcoholism.* This vitamin deficiency leads to a loss or shrinkage of neurons in many parts of the brain, especially the prefrontal cortex and parts

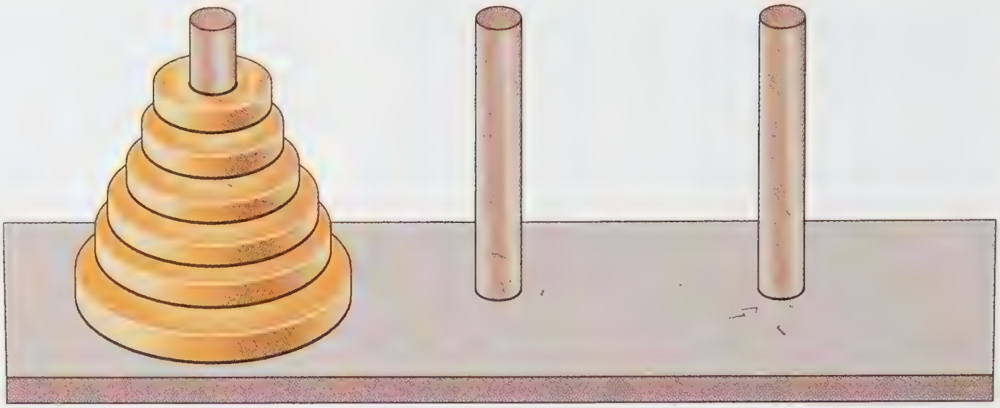


FIGURE 7.22 In the Tower of Hanoi puzzle, the task is to transfer all the disks to another peg, while moving only one at a time and never placing a larger disk onto a smaller disk. Patient H. M. learned the correct strategy and retained it from one test period to another, although he did not remember ever seeing the task before. That is, he showed procedural memory but not declarative memory.

of the thalamus. Patients suffer multiple impairments of memory, apathy, and confusion (Oscar-Berman, 1980; Squire, Amaral, & Press, 1990). Patients with Korsakoff's syndrome suffer severe retrograde amnesia, generally covering most events beginning about 15 years before the onset of their illness (Squire, Haist, & Shimamura, 1989). They also suffer from anterograde amnesia. If given a long list of words to remember, they temporarily remember the words at the end of the list, but they forget those at the beginning of the list (Stuss et al., 1994). A few minutes later, they forget even the words at the end.

Such patients have particular difficulty remembering when and where various events took place (Shimamura, Janowsky, & Squire, 1990). For example, if asked what they ate this morning or what they did last night, they give a confident but wrong answer—although it may have been true at some time in the past. In spite of their severe loss of declarative memories, they acquire new procedural memories reasonably well.

Patients with frontal-lobe damage, whatever the cause, have a characteristic pattern of answering questions with a bewildering mixture of correct information, out-of-date information, and wild guesses. Their guesses, or **confabulations,** are apparent *attempts to fill in the gaps in their memory.* Other people confabulate, too, when they cannot quite remember something that they believe they should (Moscovitch, 1995), but people with frontal-lobe damage confabulate in self-contradictory or preposterous ways, as the following example illustrates (Moscovitch, 1989, pp. 135–136):

Psychologist: How old are you?
Patient: I'm 40, 42, pardon me, 62.
Psychologist: Are you married or single?
Patient: Married.
Psychologist: How long have you been married?
Patient: About 4 months.
Psychologist: What's your wife's name?
Patient: Martha.
Psychologist: How many children do you have?
Patient: Four. (He laughs.) Not bad for 4 months.
Psychologist: How old are your children?
Patient: The eldest is 32; his name is Bob. And the youngest is 22; his name is Joe.
Psychologist: How did you get these children in 4 months?
Patient: They're adopted.
Psychologist: Who adopted them?
Patient: Martha and I.
Psychologist: Immediately after you got married you wanted to adopt these older children?
Patient: Before we were married we adopted one of them, two of them. The eldest girl Brenda and Bob, and Joe and Dina since we were married.
Psychologist: Does it all sound a little strange to you, what you are saying?
Patient: I think it is a little strange.
Psychologist: I think when I looked at your record it said that you've been married for over 30 years. Does that sound more reasonable to you if I told you that?
Patient: No.
Psychologist: Do you really believe that you have been married for 4 months?
Patient: Yes.

Why do these patients confabulate? According to Morris Moscovitch (1992), the frontal lobes are necessary for *working with memory,* the strategies we use to reconstruct memories that we cannot immediately recall. For example, if someone asks you who Romeo's girlfriend was; whether helium is a gas, liquid, or solid; or whether you have ever been to Puerto Rico, you can probably answer at once without much effort or concern. But if you are asked what you did Tuesday evening last week, or what is the farthest north that you have ever traveled, or what was the most recent time you saw any modern art, your answer will require effort and reasoning. People with frontal-lobe damage have difficulty inferring what must have happened in their past, so they make unlikely guesses.

Implicit Memory in Amnesic Patients

Brain damage can cause defects in some kinds of memory while leaving others almost intact. For example, people with amnesia have poor factual memories but can acquire new procedural memories. Moreover, these people show evidence of memories if they are tested in a special way for *implicit* memory (Reber, 1997). The kinds of memory tests we have considered so far—recall, recognition, and so forth—have been tests of **explicit memory**—*a person who states the correct answer recognizes that it is the correct answer and that it tapped his or her memory.* For example, to the question "who is your psychology instructor?" you would have to state the name or choose it from a list of choices. In contrast to such explicit or direct tests of memory, a test of **implicit** ~~*or indirect*~~ **memory** *does not require any recognition that one is using memory.* In fact, the subjects may not even realize that they are taking part in a memory test.

TRY IT YOURSELF Before we proceed, try the demonstration below: Here, you see some three-letter combinations. Add letters to form each of these into an English word:

BEL ______
HEL ______
CON ______
TRA ______
MOD ______
DEF ______

You could have thought of any number of words—the dictionary lists well over a hundred familiar CON—words alone. Did you happen to write any of the following: *below, helium, concern* or *consider* or *contrast, travel, modern, defect*? Each of these words had appeared in the two paragraphs before this demonstration. Reading or hearing a word temporarily *primes* that word and increases the chance that you will use it yourself (Graf & Mandler, 1984; Schacter, 1987). If you wrote any of these words, you probably did not realize you were using your memory, and if I had asked you to use your memory explicitly ("Please write everything you can remember from the last two paragraphs"), you still might not have included any of these words.

Brain-damaged amnesic patients show relatively normal priming, despite deficits in explicit memory. For example, one brain-damaged patient can listen to a list of words and then fill in the stems to complete them (CON-cern, TRA-vel, MOD-ern, and so forth). However, if he is asked explicitly to guess which of a pair of words was on the list (concern or contest, travel or tractor, modern or modify), he guesses with only chance accuracy. In fact, even after 40 sessions of such tests, he says he does not recognize the psychologist testing him and does not remember being tested before (Hamann & Squire, 1997).

Table 7.5 contrasts priming with several other ways of testing memory.

TABLE 7.5 Several Ways to Test Memory

	DESCRIPTION	EXAMPLE
Recall	You are asked to say what you remember.	Name the Seven Dwarfs.
Cued recall	You are given significant hints to help you remember.	Name the Seven Dwarfs. Hint: One was always smiling, one was smart, one never talked, one always seemed to have a cold. . . .
Recognition	You are asked to choose the correct item from among several items.	Which of the following were among the Seven Dwarfs: Sneezy, Sleazy, Dopey, Dippy, Hippy, Happy?
Savings (relearning)	You are asked to relearn something: If it takes you less time than when you first learned that material, some memory has persisted.	Try memorizing this list: Sleepy, Sneezy, Doc, Dopey, Grumpy, Happy, Bashful. Can you memorize it faster than this list: Sleazy, Snoopy, Duke, Dippy, Gripey, Hippy, Blushy?
Priming	You are asked to generate words, without necessarily recognizing them as memories.	You hear the story "Snow White and the Seven Dwarfs." A while later you are asked to fill in these blanks to make words: _____ L _____ _____ P _____ _____ N _____ _____ Z _____ _____ _____ C _____ O _____ E _____ _____ R _____ _____ P _____ _____ _____ P P _____ _____ A _____ H _____ U _____

CONCEPT CHECKS

11. a. Is remembering how to tie your shoes a procedural memory or a declarative memory?

b. You remember an event that happened to you the first day of high school. Is that a procedural or a declarative memory?

12. Which of the following is an example of implicit memory?

a. You read a chapter about Central America and then try to answer some practice questions about it. Although you cannot answer questions in the format "name the capital of Honduras," you do answer most of the multiple-choice items correctly.

b. Two people near you are talking about Tibet, and you are not paying attention. After they leave, someone asks you what they had been talking about and you reply that you have no idea. A few minutes later you spontaneously comment, "I wonder what's going on in Tibet these days."

13. Which kinds of memory are most impaired in H. M. and frontal-lobe patients? Which kinds are least impaired? (Check your answers on page 256.)

The Implications of Brain-Damage Amnesia

Studies of amnesic patients highlight the differences between declarative and procedural memories, between explicit and implicit memories, and between the ability to recall old memories and the ability to store new memories. The phenomenon of implicit memory demonstrates that a memory can be active and capable of influencing behavior, at least temporarily, even though a person may not be consciously aware of it.

These memory distinctions require further research, however. Many researchers are not at all convinced that psychologists can usefully distinguish between explicit and implicit memories on the basis of whether they require "conscious awareness" (Rudy & Sutherland, 1992). (Imagine trying to draw that distinction with rats' or monkeys' memories, for example.) In the future, psychologists may clarify the current distinctions or else introduce different ones. The growing consensus, however, is that some sort of distinction is necessary; we have different kinds of memory, not just one.

Infant Amnesia

Most adults remember only a few events, if any, from before age 5, although they have some memories from ages 5, 6, and beyond. Some people remember events from as far back as age 2, but those memories are isolated and fragmentary. The *relative lack of early declarative memories,* known as **infant amnesia** or **childhood amnesia,** is difficult to explain (Howe & Courage, 1997). Young children do form some reasonably long-lasting memories. Even children younger than 2 years old show clear signs of remembering events from months ago, and 4-year-olds can accurately describe experiences from ages 3, 2, and even earlier—such as their birthday parties, Christmas celebrations, and visits to grandparents (Bauer, 1996). So, it is not that young children fail to form long-term memories but that almost all of those early memories fade away.

Psychologists have proposed several theories to explain infant amnesia, none of them convincing. Sigmund Freud's theory was that infant memories are hidden or repressed because of the emotional traumas of infancy. However, he offered no evidence for this role of early trauma nor for his claim that a therapist could reaccess those memories accurately.

Another possibility is that early memories are nonverbal and later memories are verbal. One fault with that hypothesis is that 4- and 5-year-olds (who clearly rely on language) do remember events from ages 2 and 3 but older children do not.

Another possibility is that infant amnesia is related to the immaturity of the hippocampus (Moscovitch, 1985). The hippocampus is indeed slow to mature, and young children perform poorly on tests of the kinds of memory that are most dependent on the hippocampus (Overman, Pate, Moore, & Peuster, 1996). However, its development is gradual, and it evidently matures fast enough for a 4-year-old to remember events from age 2. So why can't a 10- or 20-year-old remember those same events?

Still another proposal is that a permanent memory of an experience requires the "sense of self" that develops between ages 3 and 4 (Howe & Courage, 1993). This theory, though certainly worth considering, is vague and difficult to test. Also, rats and pigeons develop long-term memories, but do we really want to say that they have a "sense of self"? In summary, the phenomenon of infant amnesia remains unexplained.

People retain many procedural memories from early childhood, such as how to eat with chopsticks or a fork and spoon, but they forget nearly all the specific events from that time of their life.

SOMETHING TO THINK ABOUT

Does the encoding specificity principle (page 245) suggest another possible explanation for infant amnesia? (Hint: Your physiological condition always differs somewhat at the time of attempted recall from what it was when originally learned.) ✱

Amnesia of Old Age

Some older people suffer from Alzheimer's disease or other degenerative conditions that gradually destroy brain cells and thereby impair attention and memory. But what about the memory of healthy older people? Years ago, psychologists tended to overstate the memory loss of old age, because they compared the average young people of their era with the average older people. The younger generation had advantages of much better nutrition, health care, and education than the older generation had ever had. Later, better research, which followed a given set of individuals over the years, found that most healthy people show little decline of memory in old age (Schaie, 1994).

On the average, older adults show only mild deficits on the simplest memory tasks, such as short-term retention of a list of words; they show greater deficits on more complex tasks (Babcock & Salthouse, 1990; Salthouse, Mitchell, Skovronek, & Babcock, 1989). If given a short narrative to remember, older adults remember the central points of the narrative almost as well as younger adults, although they are likely to forget the odd and irrelevant details (Hess, Donley, & Vandermaas, 1989).

Some old people deteriorate more than others, and we would all like to know how to increase our chances of remaining alert and productive in old age. One study found that professors at the University of California, Berkeley, deteriorated less in old age than did other people in the same region (Shimamura, Berry, Mangels, Rusting, & Jurica, 1995). The suggestion is that intellectual activity helps to protect the brain against deterioration, but we cannot draw that conclusion with confidence; it is also possible that the kinds of people least likely to deteriorate in old age are the ones most likely to pursue intellectual activities from youth on.

THE MESSAGE
The Impermanence of Memories

The various ways in which memory fails after brain damage shows us how complicated memory really is. The hippocampus and frontal cortex are clearly important for memory, but in one way or another, just about all other brain areas are, too: Memory requires sensory processing, storing information, retrieving it when necessary, reasoning about available information to fill in the gaps, and finally using the information in action. Considering all the ways that memory *could* fail, it is quite impressive that our memory is usually as accurate as it is.

SUMMARY

✱ *Normal forgetting.* Forgetting depends partly on interference from related memories. At least for memories in the phonological loop, passive decay also contributes to forgetting. We also have trouble remembering when we do not have adequate retrieval cues. (page 249)

✱ ***Amnesia after damage to the hippocampus.*** H. M. and other patients with damage to the hippocampus have great difficulty storing new declarative memories, although they form normal procedural and implicit memories. (page 250)

✱ ***Korsakoff's syndrome and other damage to the frontal lobes.*** After suffering damage to the frontal lobes, people make illogical inferences about their past and therefore make odd confabulations. (page 251)

✱ ***Lessons from amnesia.*** Studies of brain-damaged people demonstrate the value of distinguishing among different types of memory, such as declarative and procedural, or explicit and implicit. (page 254)

✱ ***Infant amnesia.*** Most people remember little from early childhood, even though preschoolers have clear recollections of experiences that happened months or even years ago. So far, psychologists have no convincing theory to explain the loss of early memories. (page 254)

✱ ***Loss of memory in old age.*** Most older people suffer some loss of memory, especially for details and for the contexts of events. (page 255)

Suggestion for Further Reading

Squire, L. R., & Butters, N. (Eds.) (1992). *Neuropsychology of Memory* (2nd ed.). New York: Guilford Press. Collection of chapters by several leading investigators of memory and amnesia.

Terms

amnesia severe loss or deterioration of memory (page 250)
hippocampus a forebrain structure in the interior of the temporal lobe that is important for storing certain kinds of memory (page 250)
anterograde amnesia the inability to store new long-term memories (page 250)
retrograde amnesia loss of memory for events that occurred before the brain damage (page 251)
procedural memory retention of learned skills (page 251)
declarative memory recall of factual information (page 251)
Korsakoff's syndrome a condition caused by a prolonged deficiency of vitamin B_1, which results in both retrograde amnesia and anterograde amnesia (page 251)

confabulations guesses made by amnesic patients to fill in the gaps in their memory (page 252)

explicit memory a memory that a person can state, generally recognizing that it is the correct answer (page 253)

implicit memory a memory that influences behavior without requiring conscious recognition that one is using a memory (page 253)

infant amnesia or **childhood amnesia** a relative lack of declarative memories from early in life (page 254)

Answers to Concept Checks

11. Remembering how to tie your shoes is a procedural memory. Remembering the first day of school is a declarative memory. (page 254)

12. Item b is an example of implicit memory. Both recall and recognition are examples of explicit memory. (page 254)

13. With H. M. and other amnesic patients, declarative memories are the most impaired and procedural memories are the least impaired. (page 254)

Web Resources

Amnesia Lab HomePage
hermes.cns.uiuc.edu/
The Amnesia Research Laboratory at the University of Illinois at Urbana-Champaign conducts neuropsychological studies of patients with memory disorders, eye movement monitoring, and functional neuroimaging. You can ask questions about amnesia that might be included in a FAQ that is being developed, although you will not receive an individual response.

The Reconstruction of Memory

Why do we sometimes report memories that turn out to be incorrect?

How do later events sometimes change people's recollections?

Forgetting has often been compared to a fading photograph, but in one important way it is very different: As we start to forget, we don't just lose the facts; we sometimes exaggerate or distort them. For example, many people who have visited India report seeing an amazing feat in which a magician throws one end of a rope into the air, where it becomes stiff long enough for a boy to climb it (Figure 7.23). Climbing an unsupported rope would be magical enough, but some people claim to have seen even more amazing feats. Researchers located 21 people who had seen the Indian rope trick and others with second-hand reports. They compared the reports to how recently the witnesses claimed to have seen the trick (Wiseman & Lamont, 1996). Here are the results:

Reported event	When seen (mean)
Saw a boy climb a rope and then climb down, but admittedly it might have been a bamboo pole instead of a rope.	4 years ago
Saw a boy start to climb a rope, then seem to vanish, then reappear at the top of the rope.	12.67 years ago
Saw a boy climb a rope, reach the top, then vanish and reappear behind the crowd.	32.5 years ago
Saw a boy climb a rope, reach the top, then vanish and reappear in a basket that had been in plain view of the audience.	41.75 years ago
Still more amazing reports...	Didn't actually see it, but heard about it from someone else.

FIGURE 7.23 Some visitors to India report seeing a boy climb a rope (or maybe a bamboo pole). Some claim they also saw him disappear and later reappear. Generally, the more spectacular the account, the less recent the event.

One possible interpretation of these results is that, over the years, Indian boys have become less magical. The more parsimonious interpretation is that, over the years, the witnesses' memories have become distorted.

Another example: You have probably heard claims that a UFO crashed near Roswell, New Mexico, in 1947, and that the U.S. government hid the spacecraft and the aliens' bodies. However, according to Kal Korff (1997), one of the most serious UFO investigators, the facts are these:

In 1947, a Roswell man reported that he might have found the remains of a flying saucer. The local Air Force commander sent Major Jesse Marcel to retrieve the wreckage. The next day, the commander announced, with Marcel present, that it was nothing but a weather balloon. (Much later, the government admitted it had actually been the remains of an experimental spying device.) No one said anything more about it until 1978—more than 30 years later—when Marcel met a UFO enthusiast and told him a version of the "wrecked saucer" story. Marcel, whose military file indicates that he had a long history of exaggerating to impress people, apparently continued that habit. For example, he bragged to the *National Enquirer* that he had shot down five enemy aircraft during the war, although his military file indicates that he was never even a pilot.

UFO enthusiasts nevertheless believed Marcel's report and boast that they then interviewed "more than ninety witnesses." However, of those, only seven claim to have actually seen anything; the other "witnesses" had just heard about it from someone else. Since then, some of those seven have changed their stories several times, contradicting themselves and one another. Some of them are apparently confusing events that they remember from the 1950s or later with the original 1947 event. In short, the parsimonious interpretation is that the memories are as distorted as those of the Indian rope trick.

Distortions of memory over time, even fairly short times, are common. They tell us much about how memory normally works and how skeptical we should be of eyewitness reports of long-ago events.

Reconstruction of the Past

If you try to recall what you did three nights ago, you will start with the details that you remember clearly and **reconstruct** the rest to fill in the gaps: *During an original experience, we construct a memory. When we try to retrieve that memory, we reconstruct an account based partly on surviving memories and partly on our expectations of what must have happened.* For example, suppose you recall that you studied in the library three nights ago. With a little effort, you may remember where you sat, what you were reading, who sat next to you, and where you went for a snack afterward. As time passes, you will probably forget that evening altogether, unless you happen to fall in love with the person who sat down next to you. If so, you will long remember that chance meeting and perhaps even where you went for a snack, but probably not which book you were reading. If you wanted to recall the book, you might reconstruct the event: "Let's see, that semester I was taking a chemistry course that took a lot of study, so maybe I was reading a chemistry book. No, wait, I remember that when we went out to eat, we talked about politics. So, I was probably reading my political science text."

We also depend on inferences to reconstruct the times of various memories. In one study, investigators located 82 adults who consistently kept up with the news and routinely watched the television program *60 Minutes.* They asked each participant to estimate the approximate dates of various news events and of various *60 Minutes* episodes. The results: People could recall the approximate dates of most news events, but their guesses about the dates of *60 Minutes* episodes were virtually worthless, except for episodes aired during the last two months (Friedman & Huttenlocher, 1997). (See Figure 7.24.) The reason for this difference is that news stories have a logical order and connections to other events in one's life. ("Oh, yeah, I remember what I did when I heard about . . .") With *60 Minutes* episodes, however, we have few links to any other experience. The important conclusion of this study is that, except for very recent events, we cannot attach a date to a memory based on its freshness or sharpness. Our attempts to date a memory involve mostly reconstructions and logical inferences.

Reconstruction and Inference in List Memory

TRY IT YOURSELF Try this demonstration: Read the words in list A, then turn away from the list and pause for a few seconds, and then write as many of the words as you can remember. Then repeat the same procedure for lists B and C. *(Please do this now, before reading the following paragraph.)*

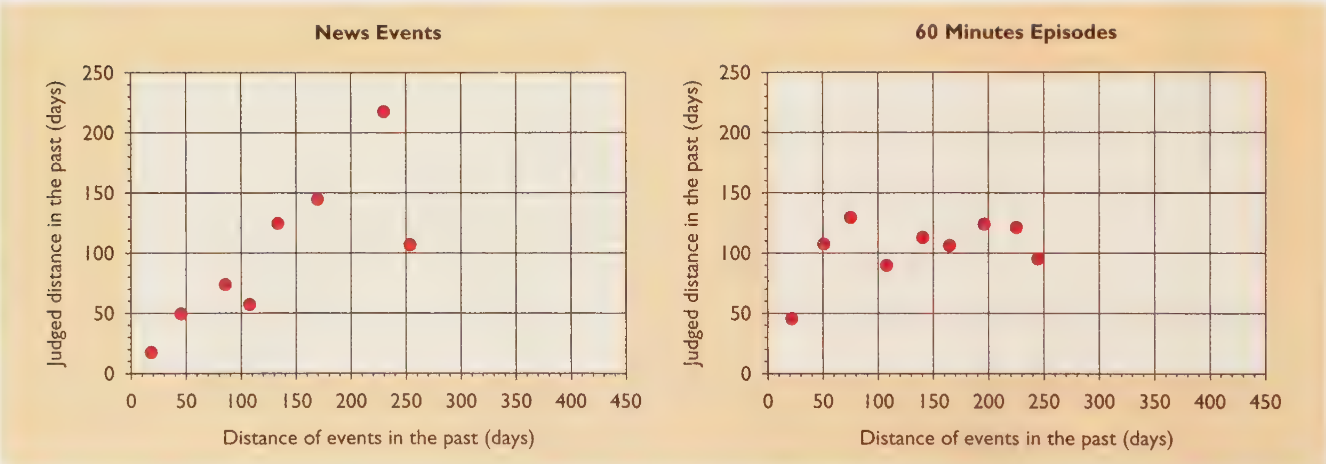


FIGURE 7.24 People who followed the news and regularly watched the television program *60 Minutes* could estimate the time of various news events but guessed almost randomly about when they saw various *60 Minutes* episodes, except those from the most recent two months. (From Friedman & Huttenlocher, 1997.)

List A	List B	List C
bed	candy	fade
rest	sour	fame
awake	sugar	face
tired	dessert	fake
dream	salty	date
wake	taste	hate
snooze	flavor	late
blanket	bitter	mate
doze	cookies	rate
slumber	fruits	
snore	chocolate	
nap	yummy	

After you have written out your lists, check how many you got right. If you missed several of them, you are normal. The point of this demonstration is not how many you got right, but whether you included *sleep* on the first list, *sweet* on the second, or *fate* on the third. Many people include one or more of these words (which were not on the lists), and some do so with great confidence (Deese, 1959). Apparently, while learning the individual words, people also learn the "gist" of what they are all about, and from that they reconstruct a memory of another word that the list implied. This effect is powerful; even adding a few unrelated words on each list does not weaken it (Robinson & Roediger, 1997). Evidently, even when we are just trying to remember a list of words, we reconstruct or infer what "must have" been on the list.

If you did include *sleep, sweet,* or *fate,* don't worry. Your error is not a sign of a bad memory; in fact, people with really bad memories *don't* make this kind of error (Schacter, Verfaellie, & Anes, 1997). They don't remember the words that *were* on the list and therefore don't use them to infer related words. If you did not include *sleep, sweet,* or *fate,* well . . . actually, this demonstration works better when people hear the list than when they read it. Try reading the lists to your roommate or another friend, and see whether they "remember" the implied words.

Reconstructions of Stories

Suppose people listen to a story about a teenager's day, including a mixture of normal events (watching television) and oddities (clutching a teddy bear and parking a bicycle in the kitchen). Which would people remember better—the normal events or the oddities? You might predict that they would remember the unusual and distinctive events, and you would be right—*if* people are tested quickly enough to show a good memory. However, if we wait long enough for people to start to forget the story, they reconstruct a more typical day for the teenager, recalling the normal events, omitting the unlikely ones, and adding some that "should have happened," even though the story did not mention them—such as "the teenager went to school in the morning." In short, the less certain people's memories are, the more they rely on their expectations (Heit, 1993; Maki, 1990).

A witness at a trial is asked to recall an event that happened months or years ago. It is neither likely nor necessary that this witness will recall every detail or every word of a conversation. Rather, the report will be a mixture of clear memories and reconstructions.

In a study that highlights the role of expectations, U.S. and Mexican adults tried to recall three stories. Some were given U.S. versions of the stories; others were given Mexican versions. (For example, in the "going out on a date" story, the Mexican version had the man's sister go along as a chaperone.) On the average, U.S. participants remembered the U.S. versions better, whereas the Mexicans remembered the Mexican versions better (Harris, Schoen, & Hensley, 1992).

Hindsight Bias

TRY IT YOURSELF

Let's try another demonstration. First read the following paragraph, and then answer the question that follows:

> For some years after the arrival of Hastings as governor-general of India, the consolidation of British power involved serious war. The first of these wars took place on the northern frontier of Bengal where the British were faced by the plundering raids of the Gurkas of Nepal. Attempts had been made to stop the raids by an exchange of lands, but the Gurkas would not give up their claims to country under British control, and Hastings decided to deal with them once and for all. The campaign began in November, 1814. It was not glorious. The Gurkas were only some 12,000 strong; but they were brave fighters, fighting in territory well-suited to their raiding tactics. The older British commanders were used to war in the plains where the enemy ran away from a resolute attack. In the mountains of Nepal it was not easy even to find the enemy. The troops and transport animals suffered from the extremes of heat and cold, and the officers learned caution only after sharp reverses. Major-General Sir D. Octerlony was the one commander to escape from these minor defeats. (Woodward, 1938, pp. 383–384)

Question In light of the information appearing in this passage, what was the probability of occurrence of each of the four possible outcomes listed below? (The probabilities should total 100%.)

a. a British victory ______%
b. a Gurka victory ______%
c. military stalemate with no peace settlement ______%
d. military stalemate with a peace settlement ______%

Note that each of these possible outcomes had some likelihood, given the facts as stated. Now that you have made your estimates of the probabilities, I can tell you what really happened: The two sides reached a military stalemate without any settlement. The British had the advantages of superior numbers and superior equipment, but the Gurkas knew the territory and refused to give up. Battles continued sporadically and indecisively for years.

Now that you know the outcome, would you like to revise your estimates of the probabilities? Perhaps if you reread the paragraph, you will decide that you had overestimated the probabilities of some outcomes and underestimated others.

Subjects in one experiment read the preceding passage about the British and the Gurkas. Some were told the correct outcome, and others were told one of the other outcomes; all were then asked to estimate the probabilities of the four possible outcomes. Each group listed a high estimated probability for the "actual" outcome they had been given (Fischhoff, 1975). That is, once they knew what "really" happened (or incorrectly *thought* they knew), they reinterpreted the previous interpretation to make that outcome seem likely, or perhaps even unavoidable (Figure 7.25). The subjects' behavior illustrates **hindsight bias**, *the tendency to mold our recollection of the past to fit how events later turned out.* Something happens and we then say, "I *knew* that was going to happen!" (Oh, incidentally, I lied about the outcome of the British-Gurkas war. Actually, the British won, although the war dragged on for 2 years. Would you like to reevaluate your estimates *again*?)

Examples of hindsight bias are abundant. As columnist Meg Greenfield (1997, p. 104) once wrote, if we believe what people say today, they always thought that "Harry Truman was a wonderful, gutsy, honest president; that the Vietnam War was a loser and we should get out; that Watergate represented a really big, serious, clearcut offense, . . . and that civil-rights legislation that upheld and enforced the principle of colorblind treatment of citizens by government was a good thing. And needless to say, everyone also always knew that the Soviet Union . . . was going to collapse and disappear." At the times of those events, however, many people who now remember thinking these things were in fact saying something quite different.

We can explain hindsight bias in several ways (Hawkins & Hastie, 1990), including that memory fades—especially

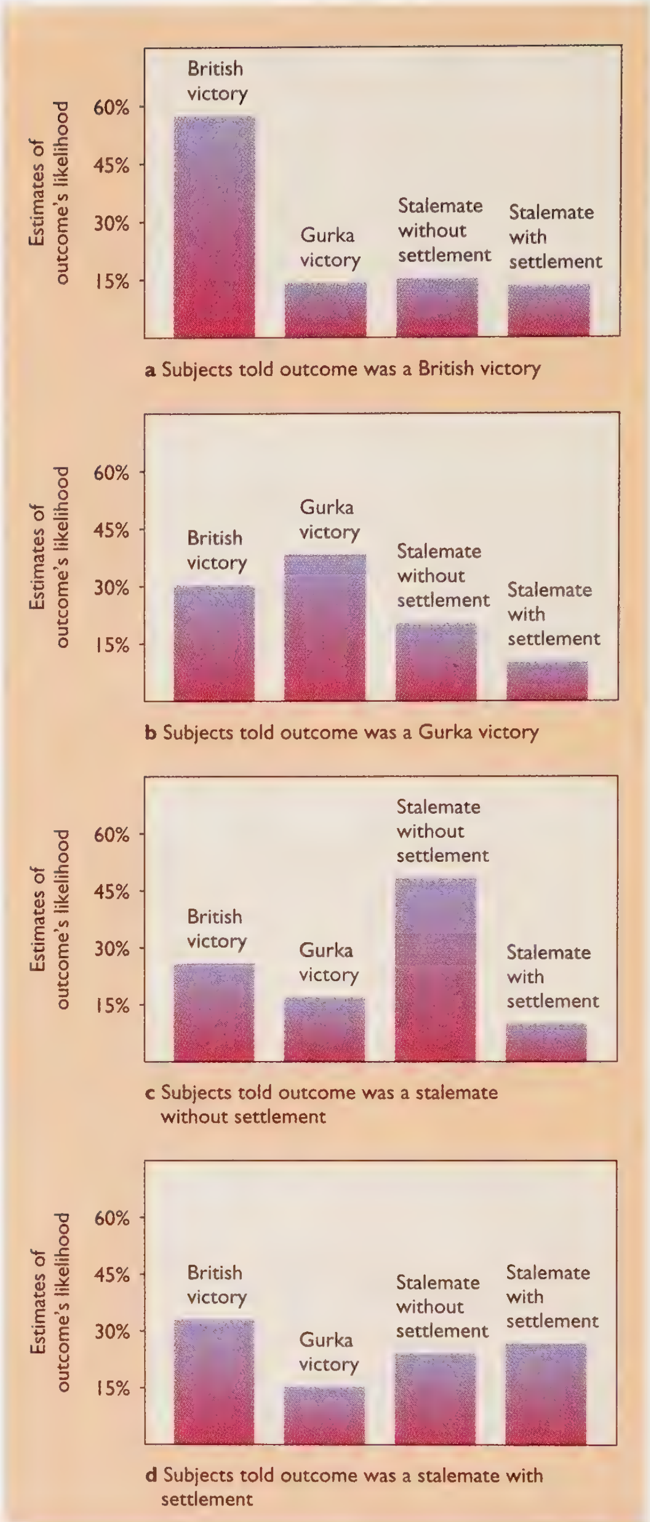


FIGURE 7.25 Mean estimates of the likelihood of four outcomes varied, depending on what each group was told about the "actual" outcome. Those who thought the British had won said that, under the circumstances, the British had a very high probability of victory. Those who thought the Gurkas had won said that was the most likely outcome under the circumstances—and so forth. (Based on data of Fischhoff, 1975.)